

DATA 606 CAPSTONE • FINAL PRESENTATION

Predicting Movie Box-Office Success Before Release

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A multi-million-dollar bet

1. Studios commit \$100M+ years before a single ticket sells
2. Reviews, word of mouth, competition — only appear after release
3. But real signals ARE known early: budget, genre, timing, studio
4. **Goal: predict financial success from pre-release info alone**

Deliverable: Data → EDA → ML models → interactive app

What counts as “success”?

$$\text{Success} = \text{Revenue} \geq 2 \times \text{Budget}$$

1. Marketing $\approx$ doubles the cost
2. Theaters keep $\sim$ half the gross
3. $\rightarrow$ must gross $\sim 2\times$ budget to break even

56%

of films clear the bar

balanced target — ideal for
classification

TMDB 5000 Movie Dataset (via Kaggle)

4,803

raw movies

3,215

after cleaning

16

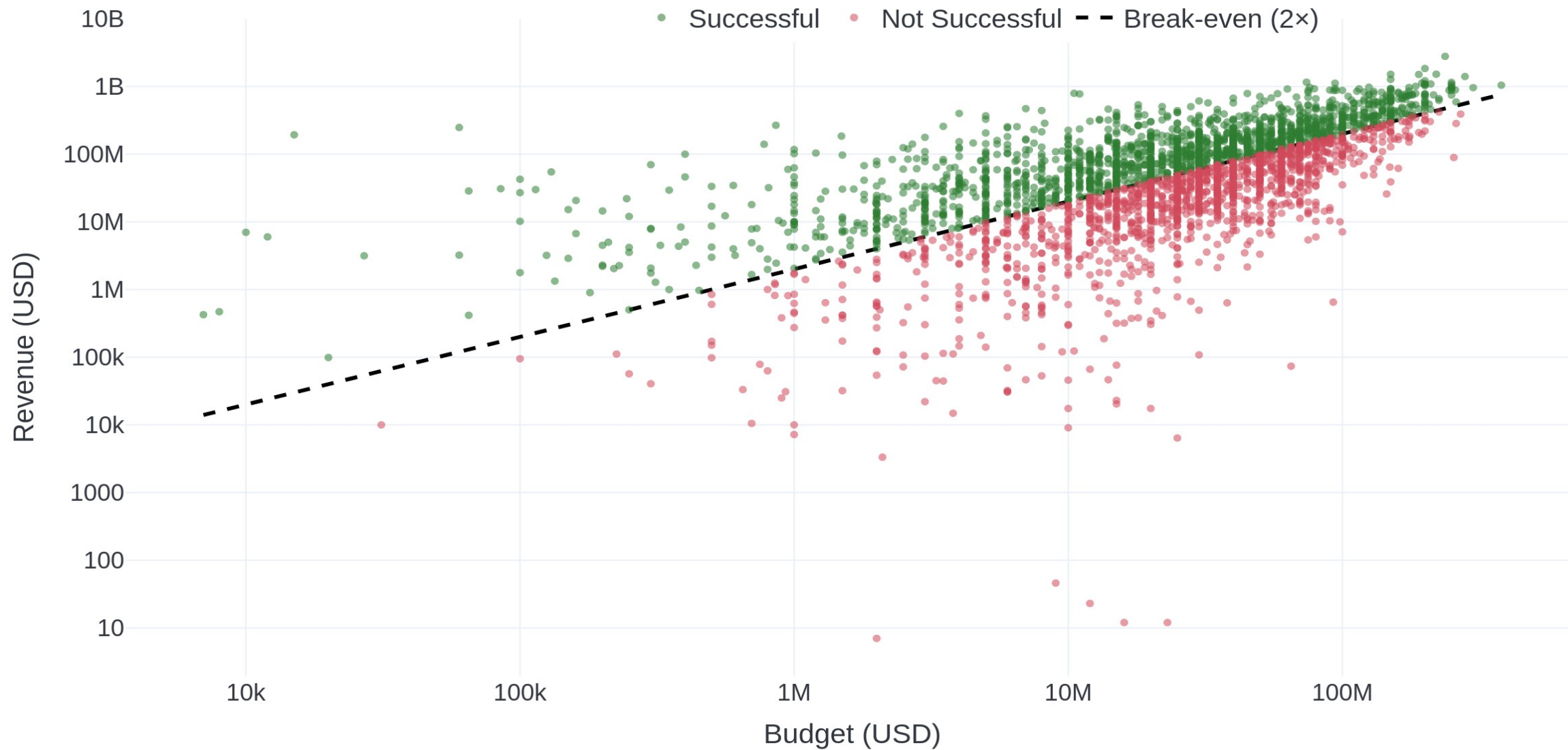
model features

**1916-
2016**

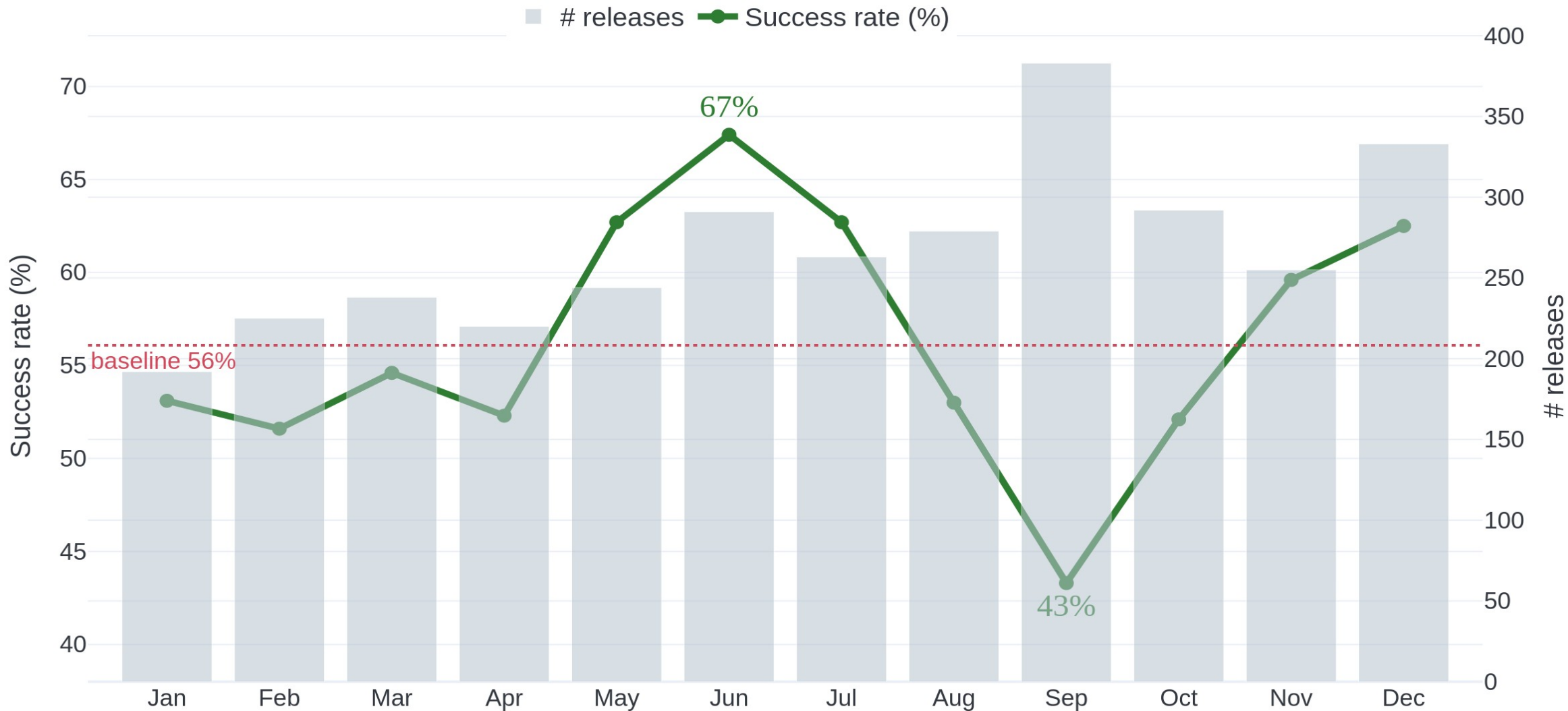
release years

1. One row = one movie (budget, revenue, genres, runtime, date, studio)
2. Dropped unknown financials + 13 impossible budgets (\$1-\$250)
3. Engineered: season · release window · sequel flag · studio flag

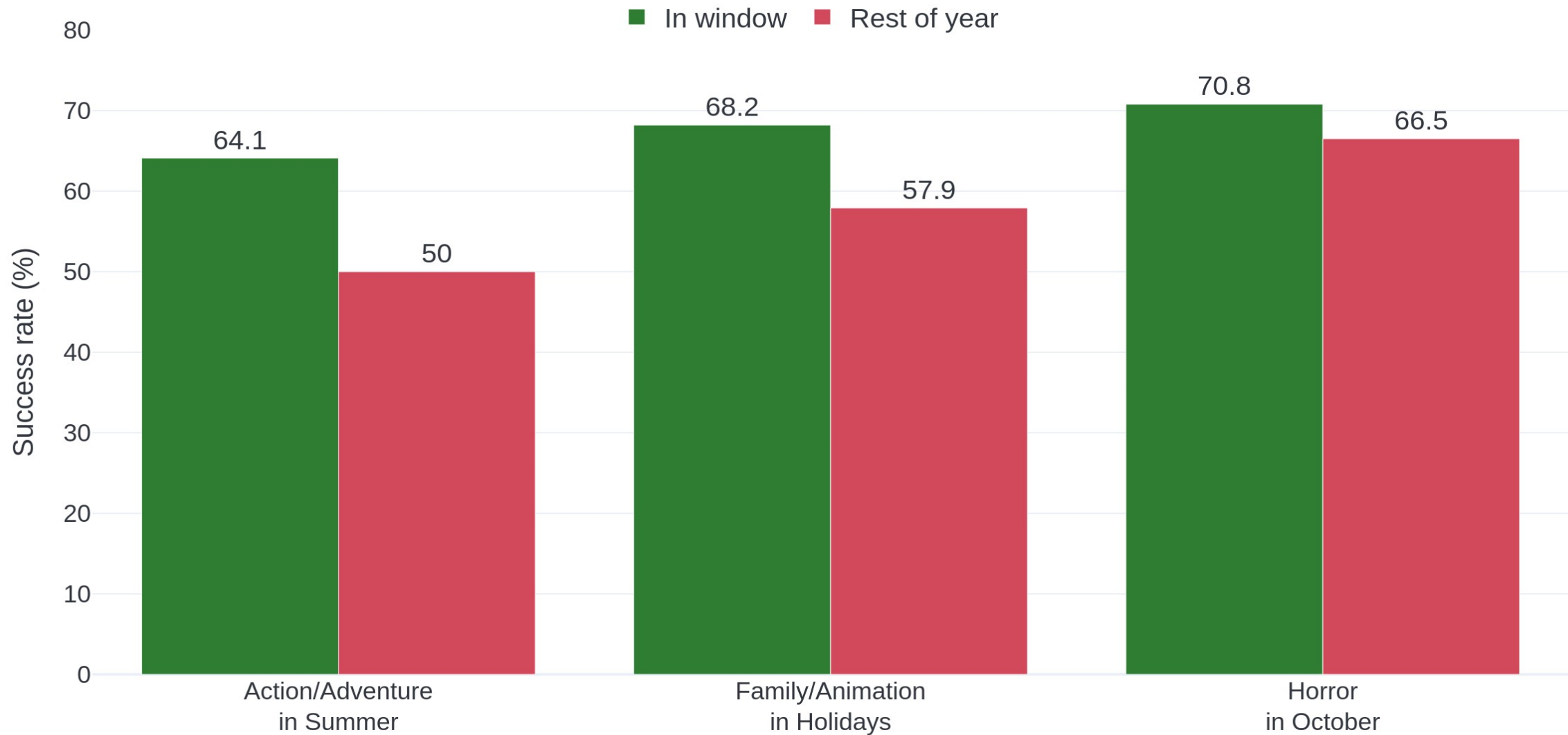
Budget vs. Revenue — the 2× break-even line



Success Rate and Release Volume by Month



The Right Genre in the Right Window



Approach — no leakage

16 features — all pre-release

- Financial: budget · spend intensity
- Content: genre · runtime · language
- Timing: year · season · window
- Production: studios · sequel flag

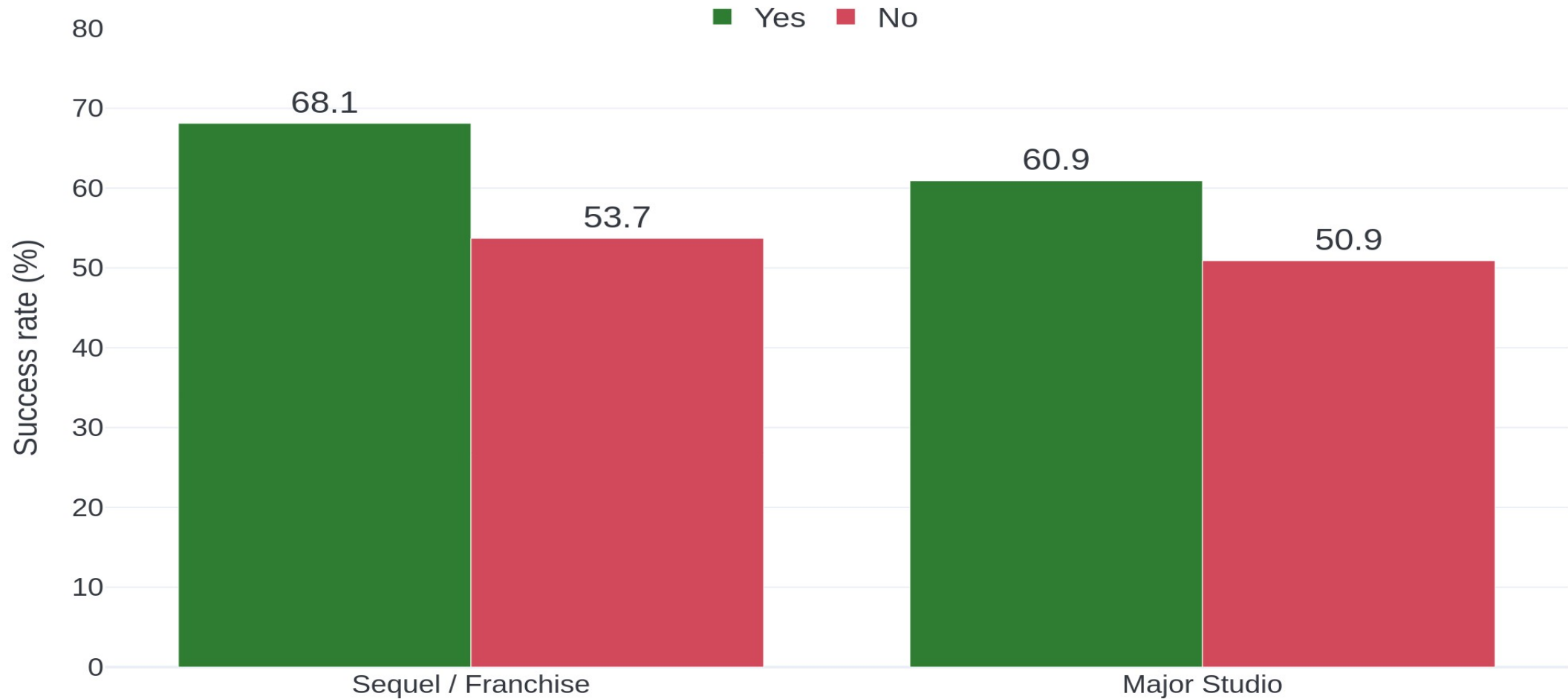
Excluded (post-release)

- Revenue · ROI · profit
- Ratings · popularity · votes
- **Model sees only what a studio knows at greenlight**

Setup: 80/20 split + 5-fold CV · 3 classifiers

MACHINE LEARNING

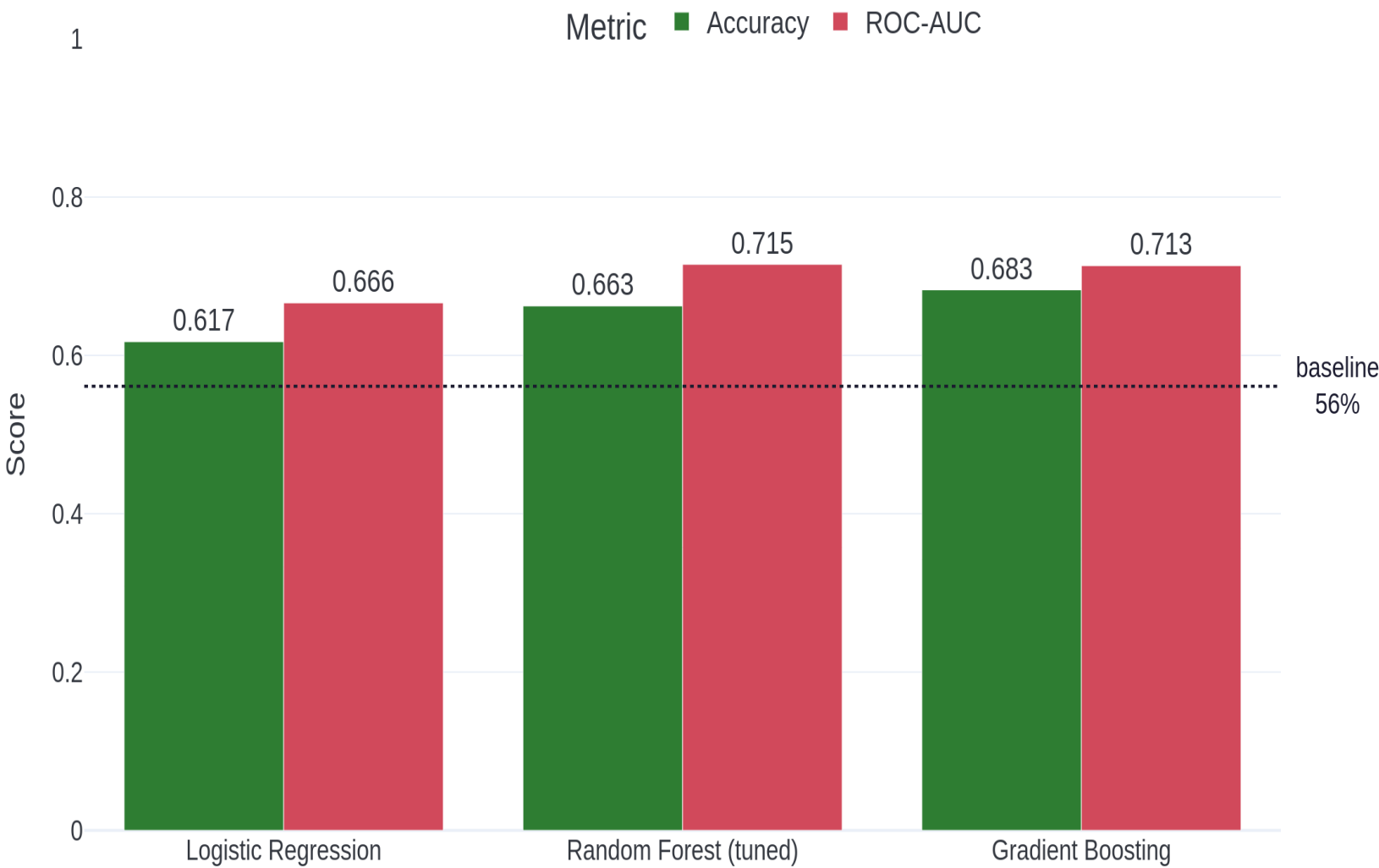
Engineered Signals: Sequel & Major Studio



Sequels: +14 pts • Major studios: +10 pts • with year + spend intensity → AUC 0.66 → 0.71

MACHINE LEARNING

Model Results — All Beat the Baseline



68%

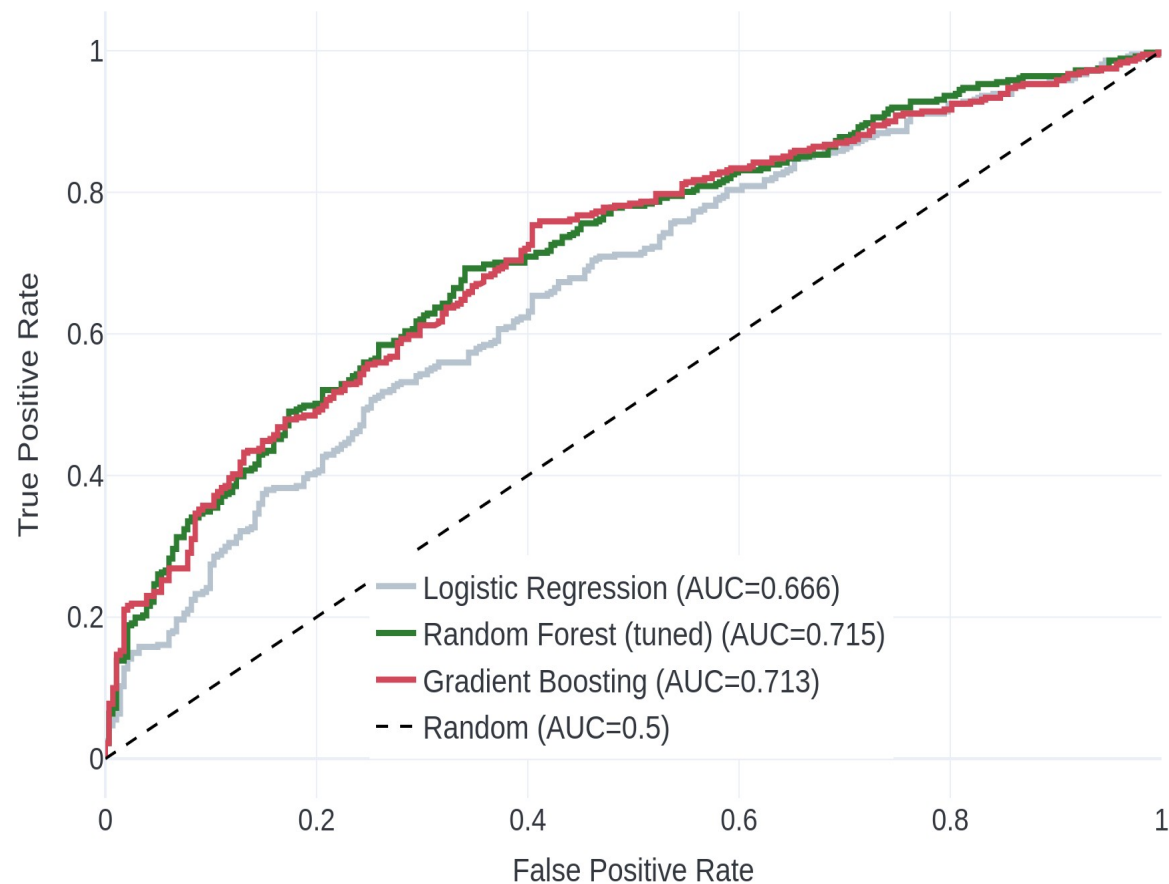
best accuracy

0.715

best ROC-AUC
(baseline 56%)

Robust across splits — CV: 64% · 0.68 AUC

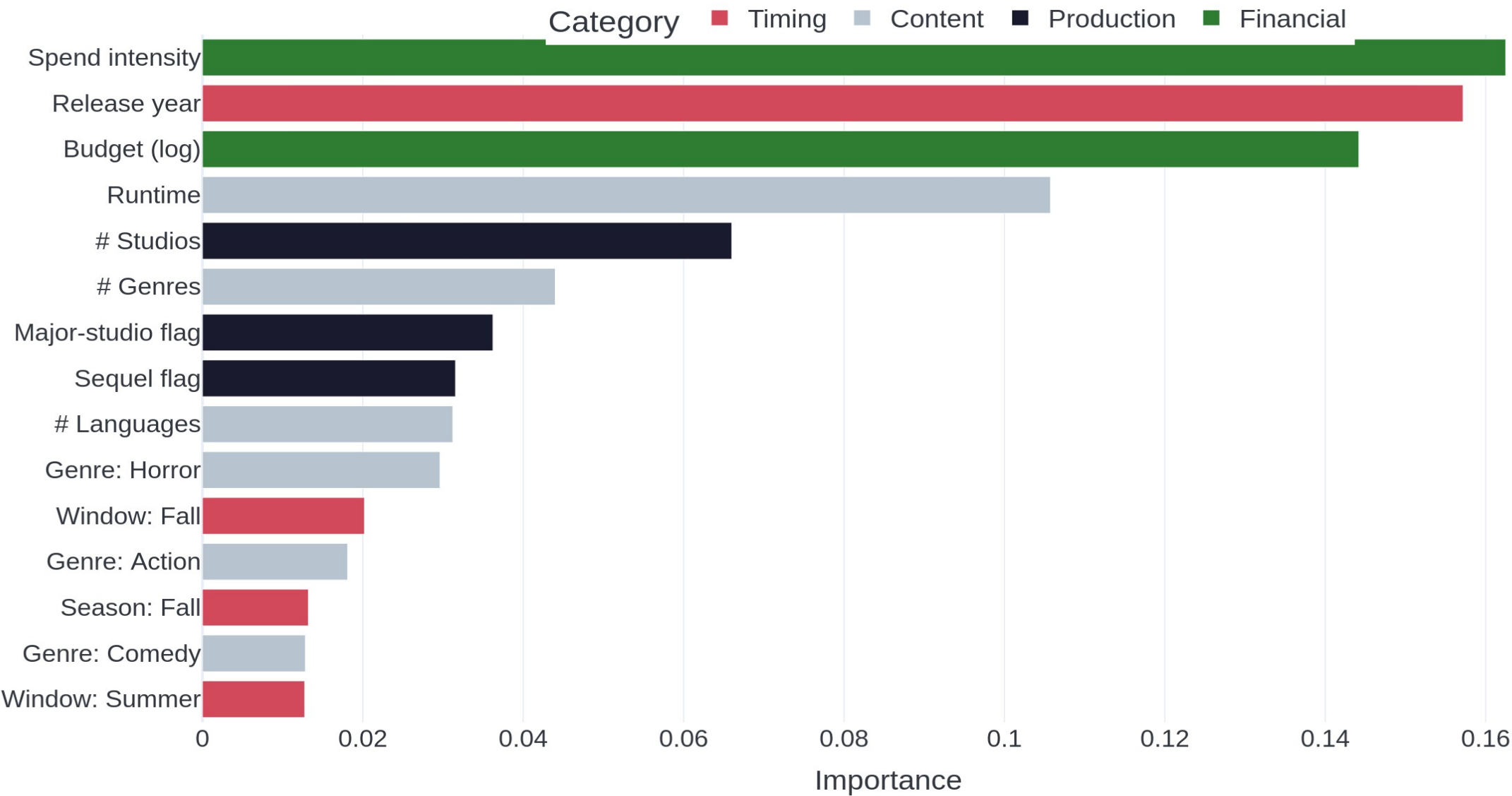
ROC Curves



Confusion Matrix — Random Forest

Actual: Not	156	126
	Pred: Not	Pred: Success
Actual: Success	91	270
	Pred: Not	Pred: Success

What Drives the Predictions — Top 15 Features



LIVE DEMO

Streamlit app · enter a film's pre-release profile → probability of success

72%

Summer franchise tentpole

72%

Low-budget October horror

48%

Big-budget September drama

Takeaways & limitations

Takeaways

- Pre-release info beats chance — clearly
- Budget dominates
- Genre + timing + franchise add real lift

Limitations

- 2× rule = proxy for profit
- Data ends in 2016, skews large
- Post-release factors excluded by design

FUTURE WORK

Where this goes next

- Star power & marketing spend as features
- Newer data + streaming-era releases
- NLP on plot summaries
- ROI tiers instead of a single 2× cutoff

Thank you — questions welcome. github.com/bdspeicher12/UMBC-DATA606-Capstone